MRI for Evaluation of Complications of Breast Augmentation

Brian D. Noreña-Rengifo, MD
Maria Paulina Sanín-Ramírez, MD
Beatriz E. Adrada, MD
Ana Beatriz Luengas, MD
Vicente Martínez de Vega, MD
Mary S. Guirguis, MD
Cristina Saldarriaga-Uribe, MD

Breast augmentation is one of the most common aesthetic procedures performed in the United States. Several techniques of breast augmentation have been developed, including the implantation of breast prostheses and the injection of autologous fat and other materials. The most common method of breast augmentation is to implant a prosthesis. There are different types of breast implants that vary in shape, composition, and the number of lumina. The rupture of breast implants is the leading cause of implant removal. The rupture rate increases substantially with the increasing age of the implant. Most implant ruptures are asymptomatic. Implant complications can be grouped into two categories: local complications in the breast and adjacent soft tissue, and systemic complications associated with rheumatologic or neurologic symptoms. The onset of local complications may be early (infection and periprosthetic collections including seromas, hematomas, or abscesses) or late (capsular contraction, implant rupture, gel bleed, or breast implant–associated anaplastic large cell lymphoma). Although mammography is the imaging modality for breast cancer screening, noncontrast breast MRI is the imaging modality of choice for evaluation of the integrity of breast implants and the complications of breast augmentation, for equivocal findings at conventional imaging, and as a supplement to mammography in patients with free injectable materials. The fifth edition of the Breast Imaging Reporting and Data System (BI-RADS) provides a systematic outline for MRI evaluation of patients with breast implants. Silicone- and water-selective sequences provide useful supplemental information to confirm intracapsular and extracapsular rupture. Breast MRI for evaluation of implant integrity does not require intravenous contrast material. The use of MRI contrast material in patients with breast augmentation is indicated when infection or malignancy is suspected. Radiologists should have a thorough understanding of the different techniques for breast augmentation, normal imaging features, and complications specific to breast augmentation.

An invited commentary by Ojeda-Fournier is available online.

Abbreviations: BIA-ALCL = breast implant–associated anaplastic large cell lymphoma, BI-RADS = Breast Imaging Reporting and Data System, FDA = U.S. Food and Drug Administration, STIR = short tau inversion-recovery

RadioGraphics 2022; 42:929–946

https://doi.org/10.1148/rg.210096

Content Codes: BR MR

From the Department of Radiology, University of Antioquia, Cra 51d #62-29, Medellín 050010, Colombia (B.D.N.R., M.P.S.R.); Department of Breast Imaging, The University of Texas MD Anderson Cancer Center, Houston, Tex (B.E.A., M.S.G.); Department of Breast Imaging, Clínica Las Américas Auna, Medellín, Colombia (A.B.L., C.S.U.); and Department of Diagnostic Imaging, Hospital Universitario Quirón Madrid, Madrid, Spain (V.M.d.V.). Presented as an education exhibit at the 2020 RSNA Annual Meeting. Received April 11, 2021; revision requested June 17 and received August 13; accepted August 16. For this journal-based SA-CME activity, the authors, editor, and reviewers have disclosed no relevant relationships. **Address correspondence to** B.D.N.R. (e-mail: *Brian.norena@udea.edu.co*).

SA-CME LEARNING OBJECTIVES

After completing this journal-based SA-CME activity, participants will be able to:

- Describe the technical characteristics of MRI for patients who have undergone breast augmentation.
- Discuss the normal findings after breast augmentation and identify the normal appearance of breast implants at MRI.
- Identify the role of MRI in evaluation for complications of breast augmentation.

See www.rsna.org/education/search/RG.

Introduction

Breast augmentation is one of the most common cosmetic surgical procedures performed in the United States. In 2019, 287 085 breast augmentation procedures were performed, 85% with silicone implants and 15% with saline implants (1). Throughout history, different substances have been used for breast augmentation, including silicone gel, hydrocarbon components, vegetable oil, and wax (2,3). In 1889, Gersunny described the injection of free substances into the breast, with complications such as paraffinomas (3). In 1917, Bartlett (4) reported on a case series in

TEACHING POINTS

- Implants are classified as retroglandular when they are placed anterior to the pectoralis muscle, or retropectoral when they are placed posterior to the pectoralis muscle.
- The onset of local complications may occur early (eg, infection or peri-implant collections including seromas, hematomas, and abscesses) or late (eg, capsular contraction, implant rupture, gel bleeding, or BIA-ALCL).
- Capsular contraction is the most common complication of breast augmentation and is a reaction to a foreign body that generates deformity and stiffness of the implant. Contracture may not be detected at imaging and, hence, is considered a clinical diagnosis.
- Intracapsular rupture occurs when the implant shell is interrupted and silicone leaks to the adjacent space, but not outside of the fibrous capsule. Extracapsular rupture is defined as a silicone leak extending beyond the fibrous capsule into the adjacent breast tissues. In extracapsular ruptures, silicone may migrate to the axillary lymph nodes and to distant locations such as the brachial plexus, upper extremities, mediastinum, liver, anterior abdominal wall, and groin.
- BIA-ALCL is seen more commonly with textured-surface implants than with smooth-surface implants. Patients present with breast swelling, pain, palpable masses, asymmetry, or lymphadenopathy. Systemic symptoms of BIA-ALCL are rare. The initial workup for BIA-ALCL is US. Breast MRI is reserved for equivocal cases.

which autologous fat grafts were used for breast reconstruction. The first reconstruction with breast implants was performed in 1895 by Vincenz Czerny, who implanted a lumbar lipoma after a breast fibroadenoma was excised (3,5). The first silicone implant was developed in 1961. Currently, most breast implants are made of either silicone or silicone combined with saline (1,6–8).

Implant rupture is one of the most common implant complications. The risk of implant rupture increases with the age of the implant. Most implant ruptures are asymptomatic, without associated trauma (9), making the diagnosis of implant rupture based on clinical findings alone difficult (10). The most common clinical finding is contour deformity of the breast (10–12). Ruptures of silicone breast implants are classified as intracapsular or extracapsular, depending on the location of the ruptured silicone with respect to the fibrous capsule. However, for saline implants, the term *deflated* is preferred (13,14). Because of its soft-tissue contrast and the ability to suppress signal, noncontrast MRI is the imaging modality of choice to evaluate the integrity of breast implants and related complications (11,15–18). In 1992, the U.S. Food and Drug Administration (FDA) prohibited the use of silicone gel–filled breast implants because of concerns that they may cause cancer, connective tissue diseases, and autoimmune diseases; however, a lack of sufficient evidence to support the relationship between implants and these diseases was subsequently shown (19,20). In 2006, the FDA recommended periodic screening with noncontrast MRI in patients with silicone breast implants to rule out subclinical implant rupture (8,21).

Radiologists must be familiar with the normal imaging findings and complications related to breast augmentation. It is important to recognize complex normal imaging findings that can simulate implant rupture. It is beyond the scope of this review to discuss findings of oncoplastic breast reconstruction procedures in which autologous myocutaneous flaps and tissue expanders are used. Furthermore, the tissue expanders are generally considered a contraindication to performing breast MRI because of the risks of overheating, expander displacement, and location marker degaussing (7,17,22,23) and the artifacts caused by the expanders, which limit the assessment of adjacent breast tissue (24) (Fig 1).

The latest edition (fifth edition) of the Breast Imaging Reporting and Data System (BI-RADS) (25) provides a systematic outline for the evaluation of breast implants with MRI and includes descriptions of implant materials and lumen types, implant locations, abnormal implant contours, peri-implant fluid, intracapsular findings, water droplets, and extracapsular silicone (22,26).

Breast Implant Classification

Implant Type

To date, five generations of implants have emerged. The fifth and latest generation of breast implant was introduced in 1993, and these implants are made of a cohesive silicone gel that allows them to maintain their contour. Also known as "gummy bear" implants, these implants have a lower incidence of implant leakage and complete collapse. The first generation of implants was made of a thick shell and gel, which led to higher rates of capsular contracture but lower rates of rupture. The second generation had a thinner shell that was filled with gel of a liquid consistency, which led to higher rates of rupture but a lower incidence of capsular contracture. The third- and fourth-generation implants had textured implant shells, leading to lower capsular contracture rates and lower rates of implant migration (16–18,22,23,27,28). There are different types, styles, and sizes of breast implants (8), including round or circular and shaped or teardrop implants. The round or circular implants are not anatomic, while the shaped or teardrop implants are taller than they are wide and are made of cohesive silicone gel (29).

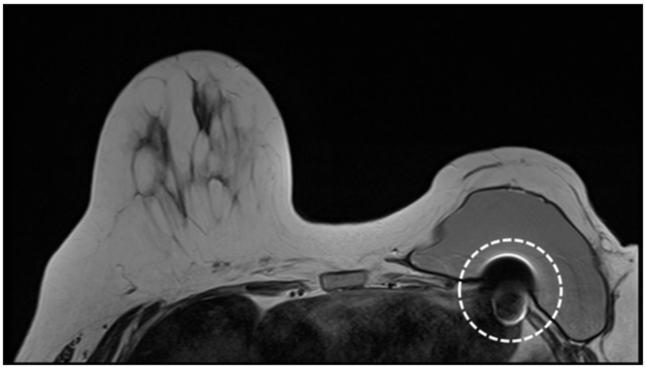


Figure 1. Tissue expander. Axial T2-weighted MR image shows a large susceptibility artifact (dashed circle) that was caused by the ferromagnetic component of the tissue expander and that resulted in incomplete visualization of the left breast and chest wall. Tissue expanders are generally considered a contraindication to performing breast MRI because of the risks of overheating, displacement of the expander, and degaussing of the location marker.

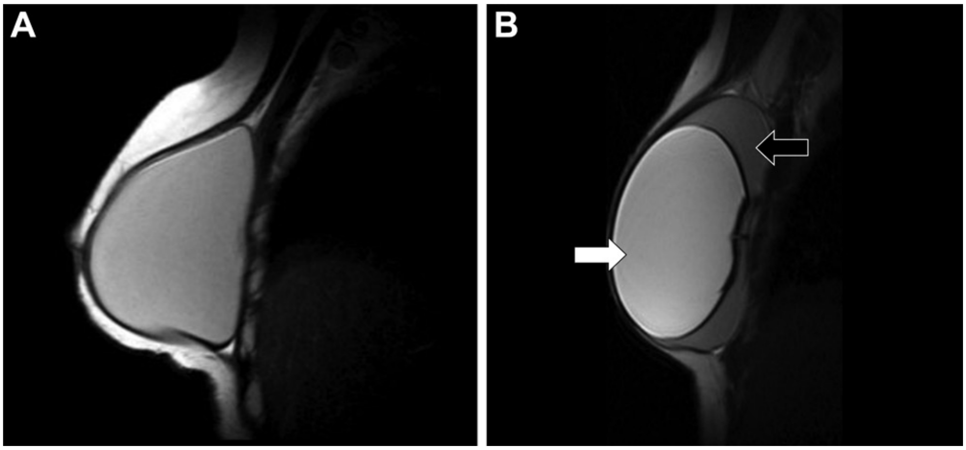


Figure 2. Sagittal T2-weighted MR images in two patients show single-lumen **(A)** and double-lumen **(B)** implants. Black arrow in **B** = external lumen, white arrow in **B** = internal lumen. Silicone shows intermediate signal intensity on T1-weighted images, high signal intensity on T2-weighted images, and signal hyperintensity on T2-weighted short tau inversion-recovery (STIR) silicone-sensitive images.

Implant Lumen

In addition to its type, an implant can also be categorized as a single-lumen or double-lumen implant. The single-lumen implant can be filled with liquid silicone gel, cohesive gel, or saline solution. The most common implants are single-lumen silicone implants. Double-lumen implants have an external lumen of saline and an internal lumen of silicone solution. Reverse double-lumen implants have an interior saline lumen and an exterior silicone lumen (15–17) (Fig 2).

Trilumen implants, which are filled with soybean oil, were previously used; however, they were banned in 1999. They had a metal chip with the implant information and showed a susceptibility artifact at imaging (7,15).

Implant Texture

The implant surface can be smooth or textured. Advantages of textured implants are decreased capsule formation and improved implant stability, with a lower likelihood of malrotation. Furthermore, textured implants are known for a lower incidence of capsular contraction. Implant texture cannot be imaged with MRI or any other modality; it is the surgeon's duty to inform the patient of the type of implant used (16–18,23,27).

Implant Location

The location of the breast implant with respect to the pectoralis major muscle is important to note. Often this can be assessed easily and reliably with sagittal MRI sequences (17). Implants are classified as retroglandular (Fig 3) when they are placed anterior to the pectoralis muscle, or retropectoral when they are placed posterior to the pectoralis muscle (16,17,22,24). Retropectoral implants are easier to displace during mammography, allowing more complete assessment of the breast parenchyma. In addition, retropectoral implants have a lower incidence of capsular contracture (24,30,31).

MRI Evaluation of the Augmented Breast and Recommended Protocol

Radiologists should be aware of the type of implant used in each patient (7,16,32). Breast MRI is an effective imaging modality to evaluate implant integrity owing to its high spatial resolution, lack of ionizing radiation, and ability to

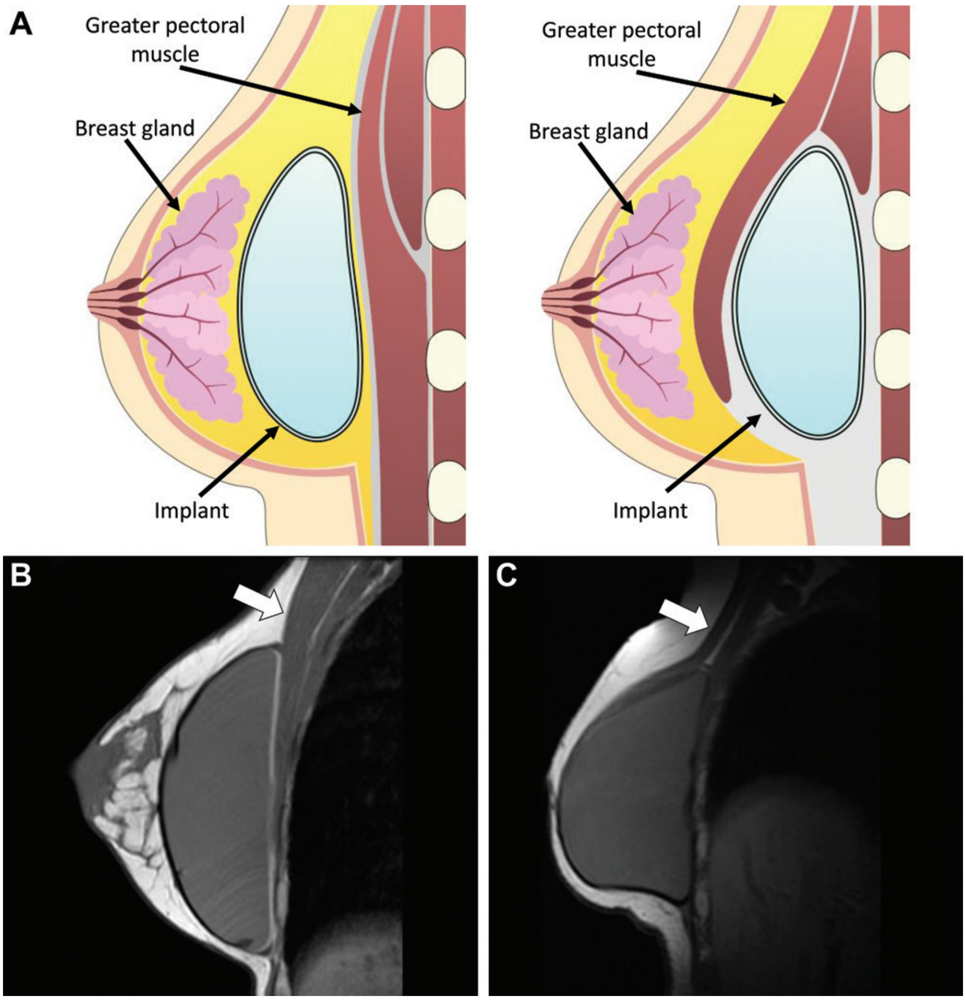


Figure 3. Breast implant position. **(A)** Illustrations show a retroglandular implant (left) lying anterior to the pectoralis muscle and a retropectoral implant (right) lying posterior to the pectoralis muscle. **(B, C)** Sagittal T1-weighted MR images show a retroglandular **(B)** and retropectoral **(C)** breast implant. Note that the pectoral muscle (arrow) is split by the implant in the retropectoral position **(C)**.

enhance or suppress signal from water, silicone, and fat (7,10,11,16,33,34). The FDA recommends using breast MRI to screen for implant rupture in asymptomatic patients with breast implants starting 5 years after implant placement and every 2–3 years after that (20,21). Breast MRI is indicated for further evaluation in patients of all ages with silicone implants and when implant complication is suspected (20). Rupture of a saline implant is a clinical diagnosis for which neither breast MRI nor any other imaging is indicated for further evaluation. Only when clinical examination is equivocal for implant rupture is imaging evaluation appropriate, according the American College of Radiology (ACR) appropriateness criteria (20).

Breast MRI is performed with the patient in the prone position, and a dedicated breast coil should be used. A magnetic strength of at least 1.5 T is required to properly evaluate tissue contrast. Fat-suppression sequences and suppression sequences or selective-excitation sequences for silicone or water are recommended (7,22). A silicone-selective sequence is a short tau inversion-recovery (STIR) sequence with selective suppression of water and fat, and a silicone-suppressed sequence involves selective suppression of silicone (7,10). Breast MRI to evaluate for implant integrity does not require intravenous contrast material (20,21). T1-weighted multiplanar MRI sequences and T2-weighted fast spin-echo sequences are both useful to evaluate silicone implants. Multiplanar MRI in the axial, sagittal, and coronal planes allows the radiologist to evaluate for radial folds that are continuous on sequential images and to differentiate them from the elastomer discontinuity of the shell, which is present when the implant is ruptured. Recommended sequences are a T2-weighted fast spin-echo sequence with a short inversion-recovery time, a water-suppression sequence (silicone hyperintense and water hypointense), a silicone-suppression sequence (silicone hypointense and water hyperintense),

Table 1: Sample MRI Protocol for Breast Implants

Sequence	Field of View (cm)	Section Thickness (mm)	Spacing	Frequency Encoding	Frequency × Phase
Axial 3D T1-weighted	28–36	1.8–2.2	. . .	Anteroposterior	384 × 384
Axial T2-weighted	32–36	3	0	Anteroposterior	320 × 256
Axial silicone specific	32–36	3	0	Anteroposterior	320 × 224
Sagittal silicone specific, left	20–24	4	0	Anteroposterior	256 × 192
Sagittal silicone specific, right	20–24	4	0	Anteroposterior	256 × 192

Note.—All axial imaging is bilateral, and sagittal imaging is unilateral. A 1.5-T imager is used for all MRI examinations at our institution. Parallel imaging is used for axial images. 3D = three-dimensional.

and a T2-weighted fast spin-echo sequence with water suppression in axial and sagittal planes (16,18,28,35).

The recommended section thickness is up to 4 mm (36). The pixel size should not be greater than 1 × 1 mm, requiring a matrix of at least 300 × 300 in a 300-mm field of view, adjusted to the patient's breast size (10,37). The use of intravenous gadolinium-based contrast material is not necessary unless a breast malignancy, breast implant–associated anaplastic large cell lymphoma (BIA-ALCL), or infection is suspected (7,22).

Other techniques have been used to determine the volume of the implant (38). Three-dimensional imaging has been used for virtual intraprosthetic navigation and evaluation of the implant capsule (39), and MRI spectroscopy is used to detect extramammary silicone; in the latter, researchers found that resonances from 0.3 to ± 0.8 ppm are attributable to silicone (40). The breast MRI protocol used at our institution in the evaluation of complications in patients who have undergone breast augmentation is summarized in Table 1.

Normal Findings at MRI of Patients with Breast Implants

Recognizing the normal findings of breast augmentation reduces the incidence of false-positive findings of implant rupture. Saline implants have a valve that is used to adjust the size of the implant. This valve is often seen at MRI as a rectangular area of signal hypointensity in the lumen. Saline implants show the signal intensity of fluid on images with all sequences. In comparison, silicone implants show high signal intensity with silicone-selective imaging sequences, and variable signal intensity on T1- and T2-weighted MR images because of the variable viscosity of the filler. Double-lumen implants (ie, an external saline component and an internal silicone component) and reverse double-lumen implants (ie, an external silicone component and an internal saline component) can be differentiated with silicone- and water-selective sequences (16,17). Special consideration should be given to breast implants that are used to correct congenital anomalies such as Poland syndrome or pectoral deformities, because they may initially appear as malpositioned implants (17). The term *stacked implants* is used when more than one implant is used for breast augmentation (28).

Radial Folds

An internal fold in the implant shell may develop when an intact implant is pressed or deformed. This may occur with both single-lumen and double-lumen implants. These folds are called radial folds and correspond to shell folds extending from the periphery of the implant. These can be found in nearly all patients with implants. In double-lumen implants, a loss of saline solution is observed over time, leading the external capsule to collapse around the silicone lumen, giving a redundant appearance that may generate extensive folds because of the contraction of the fibrous capsule (22,41).

Multiple prominent folds can simulate implant rupture, which can lead to a false-positive interpretation at MRI (15–17,34,41). Radial folds are usually thicker than the lines described as the "linguine sign" that are seen with implant rupture (41). Short and straight radial folds are called simple folds, whereas longer and curved folds are known as complex folds, which can be ramified and can mimic implant rupture (22,41). Radial folds are hypointense lines at MRI that are perpendicular to the implant's surface and can be traced up to the intact shell (Fig 4). Correlating these complex folds at multiplanar MRI is key to distinguishing them from an implant rupture. Their perpendicular orientation in comparison to the parallel orientation of an implant rupture, their sheetlike appearance, and the visualization of retained silicone inside the implant are imaging features that favor the diagnosis of a radial fold (16,41). A single image cannot allow a radial fold to be distinguished from an intracapsular rupture (24,28).

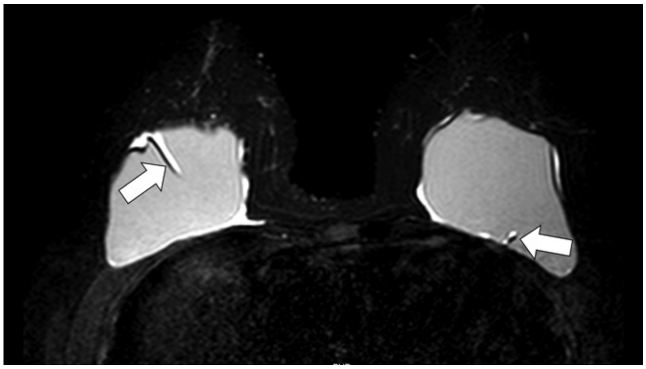


Figure 4. Normal peri-implant fluid and intact prostheses with radial folds in a 56-year-old woman. Axial T2-weighted STIR image shows curvilinear radial folds extending inward from the fibrous capsule of the implants and ending blindly in high-signal-intensity silicone (arrows), with bilateral peri-implant fluid.

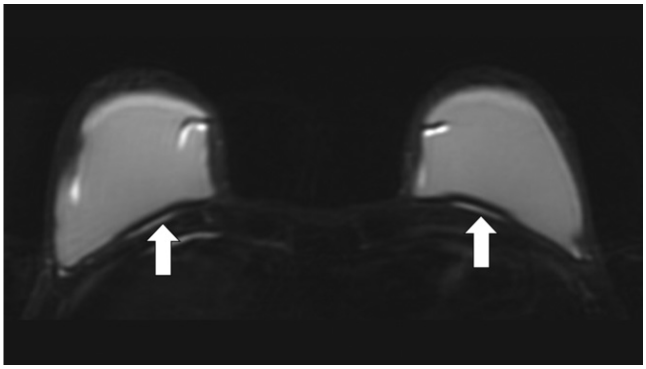


Figure 5. Normal peri-implant fluid in a 38-year-old woman. Axial T2-weighted fat-suppressed MR image shows a small amount of periprosthetic fluid (arrows) surrounding the breast implants. Axial T2-weighted STIR silicone-sensitive MR imaging did not demonstrate the fluid (not shown). These findings are consistent with periprosthetic fluid.

Peri-implant Fluid

A small amount of peri-implant fluid is a normal inflammatory response after placement of an implant (Fig 5). It is predominantly seen with textured implants and should not be confused with implant rupture. In addition, capsular thickening and adjacent calcifications are normal findings in nonruptured implants (16,22,24,28). These findings can also be associated with capsular contracture (42,43). Over time, saline implants may deflate (occurring in 1%–5% of cases over the first 2 years after placement), with the likelihood of deflation increasing over time (28). Furthermore, peri-implant fluid may also be observed surrounding saline implants, and this usually occurs because of saline migration (28).

Complications of Breast Implants

Although the life span of breast implants is approximately 10 years, common complications including rupture, displacement, and capsular contracture can occur within this time frame (19). Most complications of breast implants are not severe (44). Implant complications can be grouped into two categories: *(a)* local complications in the breast and adjacent soft-tissue and *(b)* systemic complications that are associated with rheumatologic or neurologic symptoms (45). The onset of local complications may occur early (eg, infection or peri-implant collections including seromas, hematomas, and abscesses) or late (eg, capsular contraction, implant rupture, gel bleeding, or BIA-ALCL) (Fig 6). Some complications such as herniation, asymmetry, and symmastia become more evident as the implants age (19,24,43). Symmastia occurs when both breast implants fuse together at the midline. (24) (Fig 7). Silicone implants have been associated with systemic complications or rheumatologic conditions such as Sjögren syndrome, systemic sclerosis, and rheumatoid arthritis, but no definitive cause has been reported (19,20). An association of silicone implants with melanoma and stillbirth has also been reported (19).

Peri-implant Collections

Peri-implant collections are common and are seen in 48% of patients with breast implants. They can occur in the early postoperative period and up to 1 year after implant placement. In the early postoperative period, seromas are the most common complications. Seromas are usually reabsorbed within 4–5 weeks after surgery (Fig 8). Seromas appear as fluid signal intensity on breast MR images with all sequences. Infection is a serious complication that is seen in the early postoperative period (Fig 9). Infections occur more commonly in patients who have undergone breast reconstruction than in patients who underwent breast augmentation for cosmetic reasons. The incidence is low (1%–7%). Imaging features that suggest infection are complex fluid collections, skin thickening,

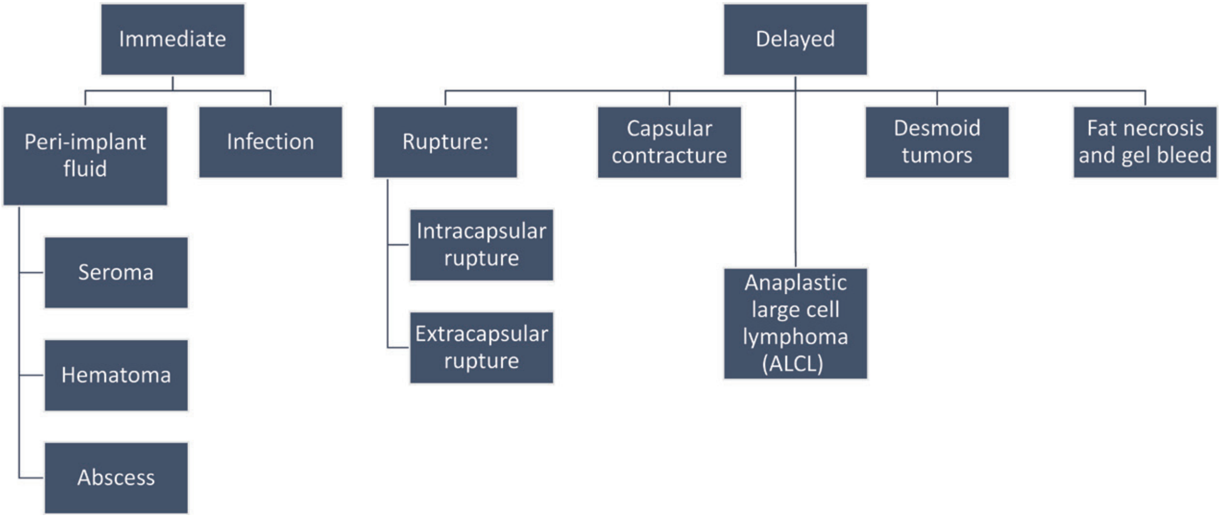


Figure 6. Diagram shows an approach to evaluating the complications of breast augmentation. Local complications may occur early (eg, infection or peri-implant fluid collections including seromas, hematomas, and abscesses) or late (eg, capsular contraction, implant rupture, gel bleed, fat necrosis, or BIA-ALCL) after implants are placed.

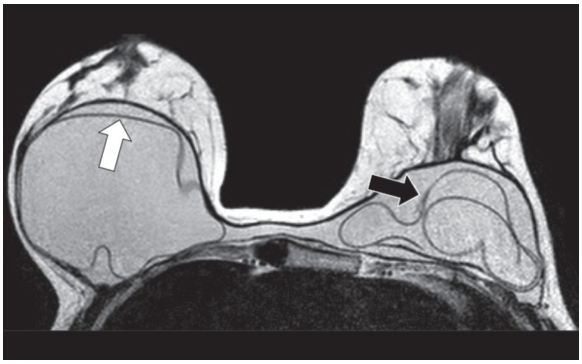


Figure 7. Symmastia in a 45-year-old woman. Axial T2-weighted MR image shows that the implant pockets have fused and the implants meet at the midline, resulting in a loss of cleavage. There are signs of bilateral intracapsular rupture, including a subcapsular line on the right (white arrow) and the linguine sign on the left (black arrow).

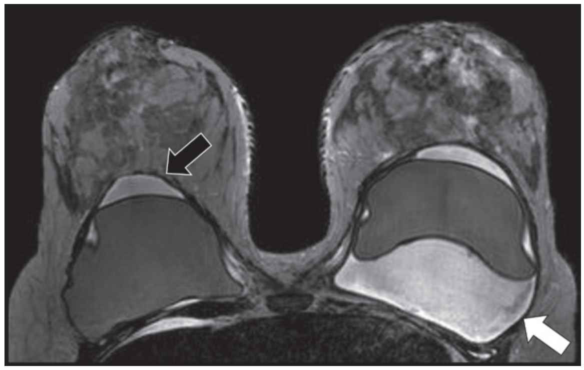


Figure 8. Early seroma in a symptomatic 40-year-old woman in the perioperative period. Axial T2-weighted MR image of bilateral single-lumen silicone breast implants shows a normal amount of peri-implant fluid in the right implant (black arrow) and a large seroma around the left implant (white arrow). The left seroma required fine-needle aspiration to remove the fluid.

edema, and capsular enhancement (6). Patients usually present with breast pain, fever, swelling, erythema, and discharge (17,23).

Hematomas occur in less than 5% of patients and usually occur in the early postoperative period. Hematomas show heterogeneous signal intensity at MRI, depending on the age of the hematoma (Fig 10), with those that develop earliest showing no enhancement and those that develop later during the 1st year after placement showing rim enhancement secondary to an inflammatory reaction (46,47).

Seromas occurring a year or more after placement of implants are considered late seromas and are rare, with a reported incidence of 1.4% (29) and 1.9% (48). They can be secondary to trauma, hematoma, infection with and without biofilm, implant rupture, synovial metaplasia, double-capsule inflammation, cancer, or BIA-ALCL (29,36,49). Most late seromas are associated with textured implants (48,49). Saline implants can be associated with fluid collections, which are thought to be related to chronic inflammation induced by the textured surface of the implant, long after implantation (>18 months) (46).

Capsular Contracture

After implantation is performed, an external fibrous capsule forms around the prosthesis (16,22). Capsular contraction occurs when the external fibrous capsule is prominent and shows abnormal contraction (16,42,43). Capsular contraction is the most common complication of breast augmentation and is a reaction to a foreign body that generates deformity and stiffness of the implant (Fig 11). Contracture may not be detected at imaging and, hence, is considered a clinical diagnosis (15,24,43,50). It may occur any

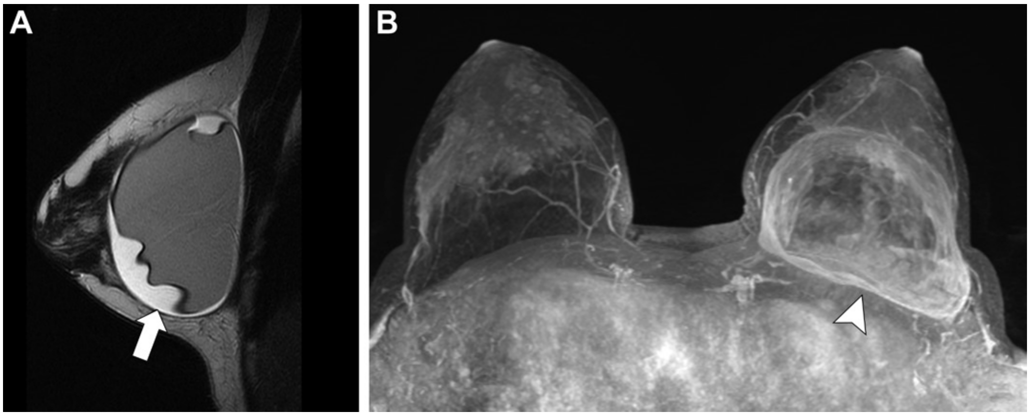


Figure 9. Left peri-implant fluid and capsulitis in a 52-year-old woman. **(A)** Sagittal T2-weighted MR image shows a collection of fluid surrounding the left retroglandular implant (arrow). **(B)** Axial maximum intensity projection reconstruction MR image shows asymmetric capsular enhancement, which is more evident on the left side (arrowhead); these findings are associated with an exacerbated inflammatory reaction of the capsule or capsulitis. The patient received antibiotic therapy and showed improvement.

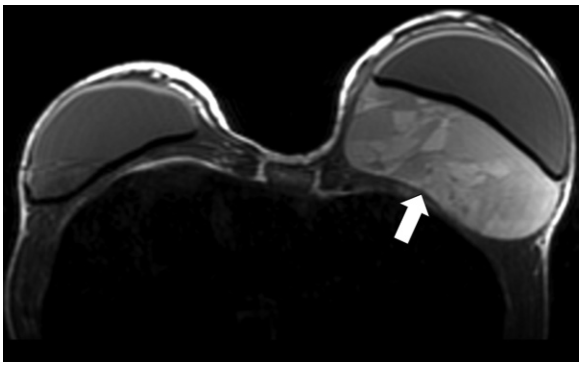


Figure 10. Left periprosthetic hematoma after breast trauma in a 35-year-old woman. T1-weighted MR image shows a large hyperintense collection of fluid that displaces the left breast implant anteriorly. Surgical drainage of the hematoma was necessary.

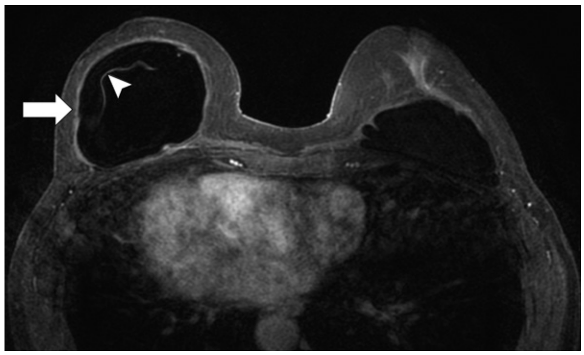


Figure 11. Capsular contracture in the right implant in a 42-year-old woman. Axial T1-weighted MR image shows the increased anterior-posterior diameter of the right implant in comparison to the normal left implant. Arrow = asymmetrical capsular enhancement and thickening, arrowhead = radial folds.

time after placement of the implant, but it usually occurs in the first few postoperative months (42). A higher incidence rate has been reported with first- and fourth-generation implants (23) than with those of other generations and with silicone implants than with saline implants (24). Contracture occurs more frequently with smooth-surface silicone implants than with textured implants (42,48). It also occurs more frequently in retroglandular implants (8.6%) than in retropectoral implants (2.8%) (42,51). Radiation therapy may promote capsular contracture (52).

The Baker staging system divides capsular contraction into four grades. The Baker classification is summarized in Table 2 (53). Handel (54) reported that grade 2 contraction occurs most frequently (40%), followed by grade 1 (30%), grade 3 (22%), and finally, grade 4 (8%). Although it is largely a clinical diagnosis, irregular thickening of the fibrous capsule can be identified at breast MRI. In addition, the implant can become spherical (ie, an increased anteroposterior diameter) and firm. There is also an increase in the number of radial folds (16,24,54,55). Of note, capsular enhancement can be observed in asymptomatic patients and should not be interpreted as abnormal (23).

Table 2: Baker Classification of Capsular Contracture

Grade	Description
Grade 1	Normal implant with normal smooth texture
Grade 2	Implant with normal contour and firm texture
Grade 3	Implant with firm texture and contour alteration
Grade 4	Breast is painful as a result of the capsular contracture

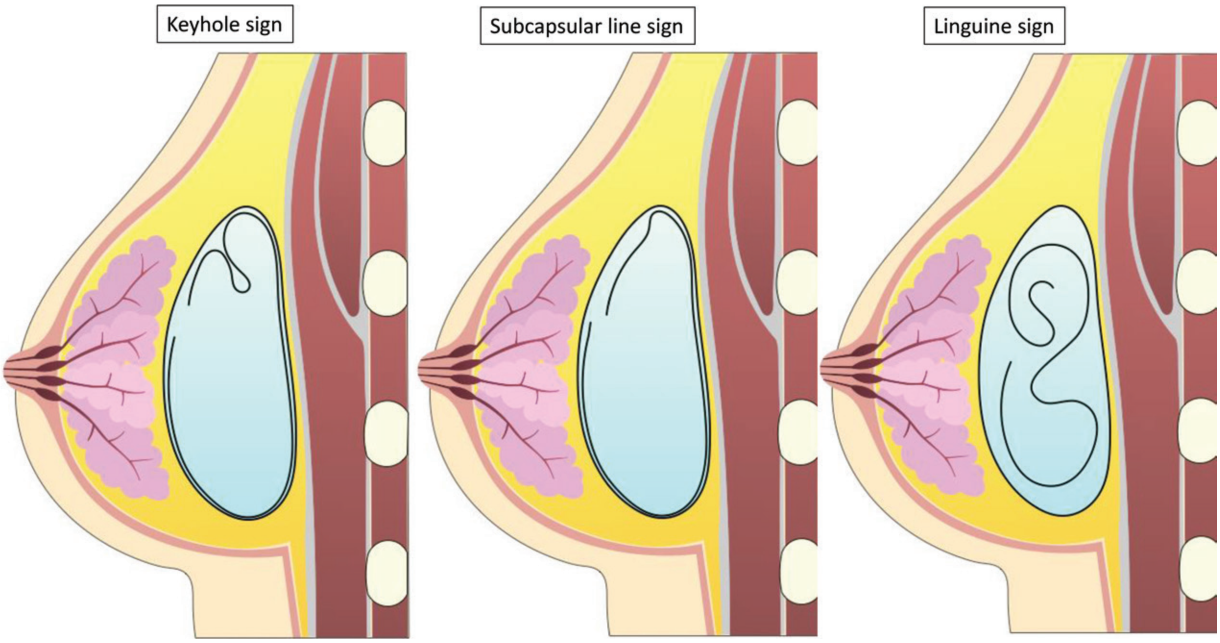


Figure 12. Findings of intracapsular rupture at MRI. Illustrations show the keyhole sign (left), with silicone expanding the vertex of the radial fold; a subcapsular line (middle), with lines running parallel to the fibrous capsule; and the linguine sign (right), with folded wavy multidirectional lines within the silicone gel, representing the collapsed implant shell.

Implant Rupture

Clinical diagnosis of implant rupture is difficult, and it can be missed in more than one-half of cases (16,22). MRI is the diagnostic test of choice to evaluate for implant rupture and is currently the imaging method recommended by the FDA (7,16,20,36,56). Implant rupture can be intracapsular or extracapsular, with intracapsular ruptures representing most cases of implant rupture (78% vs 22%) (16,18,43).

Intracapsular rupture occurs when free silicone emerges from a tear in the elastomer shell but the fibrous capsule remains intact, whereas extracapsular rupture occurs when the free silicone enters the surrounding mammary parenchyma secondary to a breakdown of the elastomer shell and the fibrous capsule (11,18). Implant rupture can occur spontaneously or can be induced by trauma (22,57). MRI is the best modality to identify the presence and extent of a silicone leak (16).

Although the exact incidence of breast implant rupture is unknown (43), approximately 50% of implants have ruptured at 10 years after implantation (11,18,37), and most ruptures occur 10–15 years after implantation (9,58). Capsular rupture applies to single-lumen or double-lumen silicone implants. The saline content of saline implants and double-lumen implants is rapidly absorbed when these implants rupture (24,43). Pain and decreased breast size are diagnostic, and imaging is not required. *Implant deflation* is the preferred term for saline implants, instead of rupture (14,24,45).

MRI has high sensitivity (64%–100%) and lower specificity (63%–97%) for determination of implant rupture (10,11,18,32,45,57,59), with accuracy of 92%, a positive predictive value of 50%–100%, and a negative predictive value of 70%–100% (10,32,59). MRI accuracy for detection of rupture depends on the type of implant (59). Mammography has sensitivity and specificity of 20% and 89%, and US, 30% and 81%, respectively (57). Maijers et al (60) evaluated the interobserver variability at MRI and found a κ coefficient of 0.92 and 0.74 for diagnosis of intracapsular and extracapsular rupture, respectively. Intracapsular rupture occurs when the implant shell is interrupted and silicone leaks to the adjacent space, but not outside of the fibrous capsule. Extracapsular rupture is defined as a silicone leak extending beyond the fibrous capsule into the adjacent breast tissues (16,22). In extracapsular ruptures, silicone may migrate to the axillary lymph nodes and to distant locations such as the brachial plexus, upper extremities, mediastinum, liver, anterior abdominal wall, and groin (22,61).

Only the "keyhole" sign, the "linguine" sign, and the subcapsular line are included in the BI-RADS lexicon to describe intracapsular rupture (25); however, other signs that could be included in future editions of BI-RADS have been described. Signs of capsular rupture can be grouped into signs of possible rupture and definitive signs of rupture (22) (Figs 12–14). Signs of possible rupture include *(a)* contour deformity including contour bulging, called the "rat tail" sign,

Figure 13. Signs of possible intracapsular rupture in two patients. **(A, B)** Sagittal T2-weighted STIR silicone-selective MR image **(A)** in a 37-year-old woman shows the teardrop sign (arrow). Axial T1-weighted MR image **(B)** shows the keyhole sign (arrow). **(C)** Sagittal T2-weighted MR image in a 44-year-old woman shows the water droplets that indicate the "salad oil" sign (white arrow), with a definitive linguine sign (black arrow), indicating intracapsular rupture.

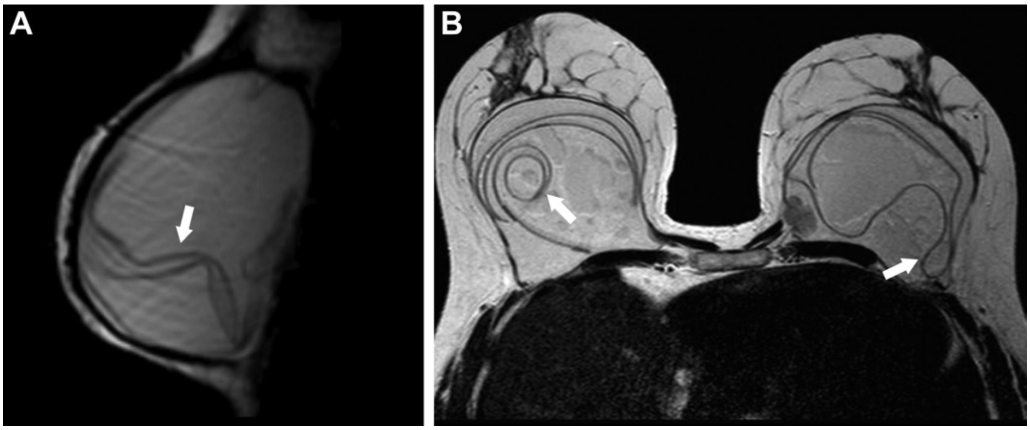


Figure 14. Signs of definitive intracapsular implant rupture in a 46-year-old woman. **(A)** Sagittal T2-weighted MR image shows subcapsular lines (arrow) running parallel to the fibrous capsule. **(B)** Axial T2-weighted MR image shows the linguine sign in both prostheses (arrows).

when a thin silicone border extends to the chest wall (16,32); *(b)* irregular borders usually with calcifications from the fibrous capsule, which looks like an indistinct border; *(c)* "salad oil" sign or "droplet" sign, when silicone gel is mixed with reactive subcapsular fluid; *(d)* "keyhole" sign, in which the implant shell invaginates and the membranes do not touch each other, which is caused by gel bleeding or silicone outside the shell of the implant; and *(e)* "teardrop" sign, in which an implant's shell invagination contains a silicone drop because of a silicone leak through a small tear (6,15,16,24,34,61). These signs could be nonspecific for the diagnosis of intracapsular rupture and could lead to wrong diagnoses due to false-positive results (12,23), but the radiologist can use the linguine sign to diagnose rupture accurately (61). False-positive results of the linguine sign are due to the presence of atypical or complex folds (23,32).

The definitive signs of rupture include *(a) subcapsular lines*, which are defined as hypointense lines surrounded by silicone gel that run parallel to the fibrous capsule (when the lines are paired and running parallel to the articular capsule, they are known as the "railroad track" sign and are considered an early form of the linguine sign); *(b)* free silicone or silicone granulomas, indicating rupture of the shell and capsule, with extravasation to adjacent tissue (Fig 15); and *(c)* the linguine sign, in which low-signal-intensity curve lines inside the silicone represent collapsed elastomer floating inside the silicone and contained by the fibrous capsule (16,22,41,43,57). The linguine sign is the most sensitive and specific sign. Vestito et al (62) reported sensitivity of 96% and specificity of 77% (32,43). Furthermore, the most reliable sign to diagnose extracapsular rupture is the presence of free silicone in the breast parenchyma or the axillary lymph nodes in the presence of intracapsular rupture (43,57). Free silicone shows isointensity or low signal intensity at T1-weighted MRI with fat suppression and

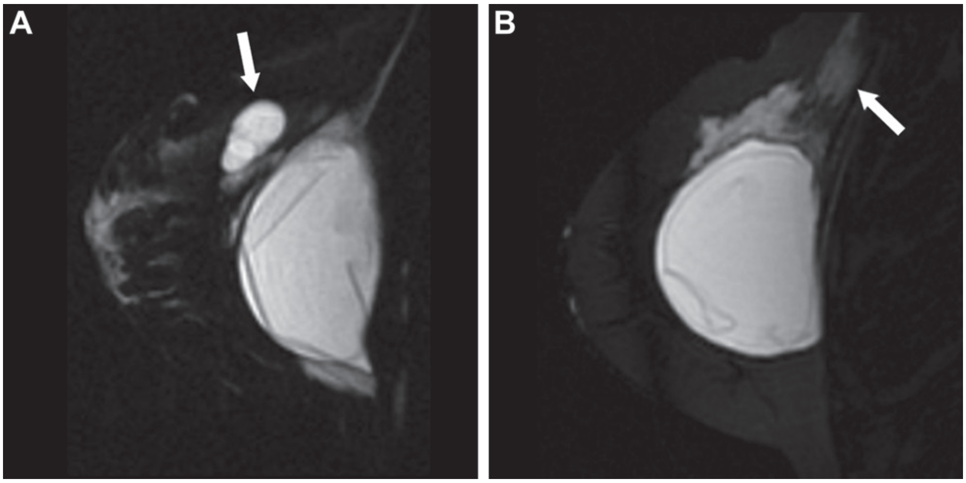


Figure 15. Signs of bilateral intracapsular and extracapsular rupture in a 53-year-old woman. **(A)** Sagittal silicone-selective MR image of the right breast shows extrusion of silicone into the adjacent soft tissue, which has resulted in a silicone granuloma (arrow). **(B)** Sagittal silicone-selective MR image of the left breast shows free silicone infiltrating the pectoralis muscle (arrow), related to extracapsular rupture of the left implant. In addition, there were signs of bilateral intracapsular rupture and left silicone-laden lymphadenopathy (not shown).

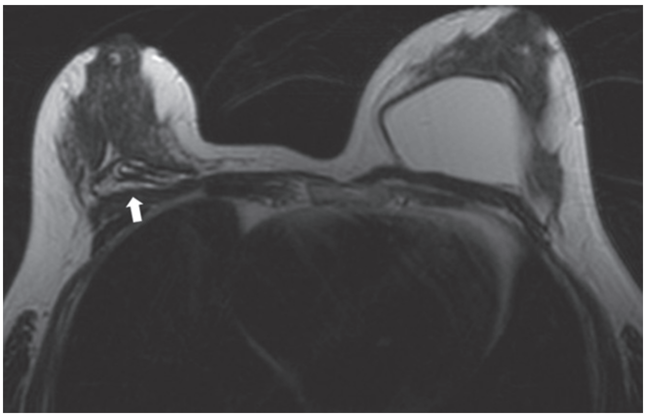


Figure 16. Collapse of a saline breast implant in a 56-year-old woman with a clinical change in the shape of the right breast. Axial T2-weighted MR image shows complete deflation of the right saline implant (arrow). Implant deflation is a clinical diagnosis and requires no MRI examination.

high signal intensity at T2-weighted MRI with water suppression. Silicone granulomas may show enhancement similar to that of a breast carcinoma and may require biopsy to exclude malignancy (43). Relying on only one sign, such as the linguine sign, without other supporting findings of implant rupture can result in a false-positive diagnosis of implant rupture (62).

Maijers et al (60) developed a system called SI-RADS (Silicone Implants Reporting and Data System); however, it is not one of the ACR RADS systems. This system uses a score of 0–4 to define intracapsular or extracapsular rupture and offers respective treatment recommendations. A score of 0 requires additional imaging or a second opinion from a colleague radiologist, a score of 1–2 does not require any clinical treatment, and a score of 3–4 requires referral to a plastic surgeon (60). In addition, Middleton (14) defined a staging system for intracapsular rupture as uncollapsed (grade I), minimally collapsed (grade II), partially collapsed (grade III), and fully collapsed (grade IV) (Fig 16). Potential pitfalls of imaging implant rupture are described in Table 3 (Figs 17–19).

Deformities, Displacement, and Malposition of the Implant

Capsular herniation generates a contour irregularity, with protrusion of at least 1 cm, which may or may not be associated with implant rupture and may be a variation of the fibrous capsule shape that surrounds an intact implant (45,46) (Fig 20). Grouped folds in the herniated portion can be observed at MRI (45). Breast implants may be subject to displacement, which is an indication for surgical removal or replacement (15,62) (Fig 21).

Desmoid Tumors

Also known as fibromatosis, desmoid tumors are rare tumors that are known for fibroblastic proliferation (10,60). Desmoid tumors may be

Table 3: Pitfalls of Imaging Implant Ruptures

Pitfall	Details
Radial folds	Radial folds may be misinterpreted as signs of intracapsular rupture (41,45). Imaging features that favor a diagnosis of implant folds are their perpendicular orientation, a sheetlike appearance, and the confirmed presence of retained silicone inside the implant (16,41).
Gel bleed	A gel bleed is a microscopic silicone leak through an intact capsule that may explain the presence of extracapsular silicone in axillary lymph nodes or distant sites (22,43). A gel bleed is difficult to identify unless it is extensive, because the migration of silicone is microscopic. A large gel bleed shows the keyhole sign at imaging (Fig 17A) (15).
Water droplets	Water droplets or small amounts of air in a silicone implant are not reliable signs of rupture of the prosthesis (Fig 17B) (34). Water droplets may be visualized in the silicone lumen, commonly after injection of steroids, betadine, or antibiotics (23).
Previous implant rupture	Distant silicone may be secondary to extracapsular rupture of a previous implant (Fig 18). Therefore, the patient's history of previous implants is important. In these patients, no other signs of intracapsular rupture are present (15,22).
Double-lumen implants	Double-lumen implants may imitate an intracapsular rupture, because the appearance may mimic the linguine sign (32). The distinction of double-lumen implants from single-lumen implants that show a reactive effusion is important. It is also important to consider that a double-lumen implant with a deflated saline compartment can resemble a single-lumen implant (23). Adjacent hematomas or seromas may also simulate implant ruptures; however, they can be diagnosed on the basis of signal intensity with MRI (45).
MRI technique artifacts	Ghost artifacts due to movement may mimic the subcapsular line sign; incomplete water suppression at silicone-sensitive MRI may generate fluid or cystic lesions to simulate extracapsular silicone; and the chemical shift artifact in the silicone-to–soft-tissue interface may simulate encapsulation (23,61).
Acellular dermal matrix	Acellular dermal matrix (AlloDerm, LifeCell), which is sometimes used in reconstruction with breast implants, can be confused with an extracapsular implant rupture, a hematoma, or an abscess. AlloDerm may be difficult to identify without viewing the surgical description (63) (Fig 19).

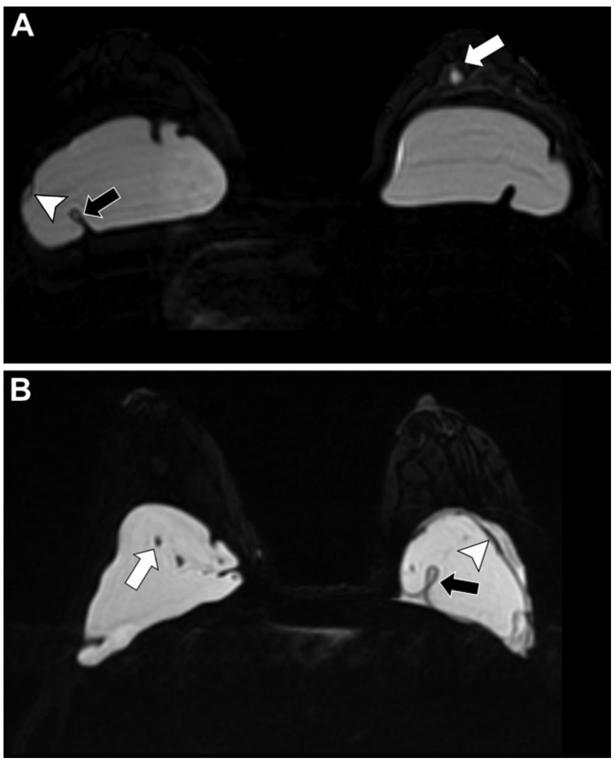


Figure 17. Implant rupture pitfalls. **(A)** Gel bleed in an intact silicone implant in a 52-year-old woman. Axial bilateral silicone-sensitive MR image shows high signal intensity anterior to the intact left implant (white arrow), indicative of a gel bleed due to microscopic diffusion of silicone gel through the breast implant shell. In some cases, this silicone can migrate to regional lymph nodes. Signs of intracapsular rupture are present in the right implant, including the keyhole sign (black arrow) and a subcapsular line (arrowhead). **(B)** Axial bilateral silicone-sensitive MR image in a 46-year-old woman shows water droplets in the intact right prosthesis (white arrow) and signs of intracapsular rupture in the left prosthesis that are defined by the keyhole sign (black arrow) and a subcapsular line (arrowhead). Visualization of water droplets or a small amount of air inside a silicone implant, with no other evidence of rupture, is not a sign of implant rupture.

indolent tumors, or they may have an aggressive course that mimics that of breast carcinomas (10) (Fig 22). These tumors are associated with trauma, scarring, or exposure to hormones. They have also been shown to have a relationship with implant capsules (10), although the etiology is not yet clear. In a review (64) of 32 published cases, 17 were associated with silicone implants, and the mean time of onset after implant placement was 3.1 years. Desmoid tumors are usually intramuscular. Breast MRI is the best imaging modality to evaluate for chest

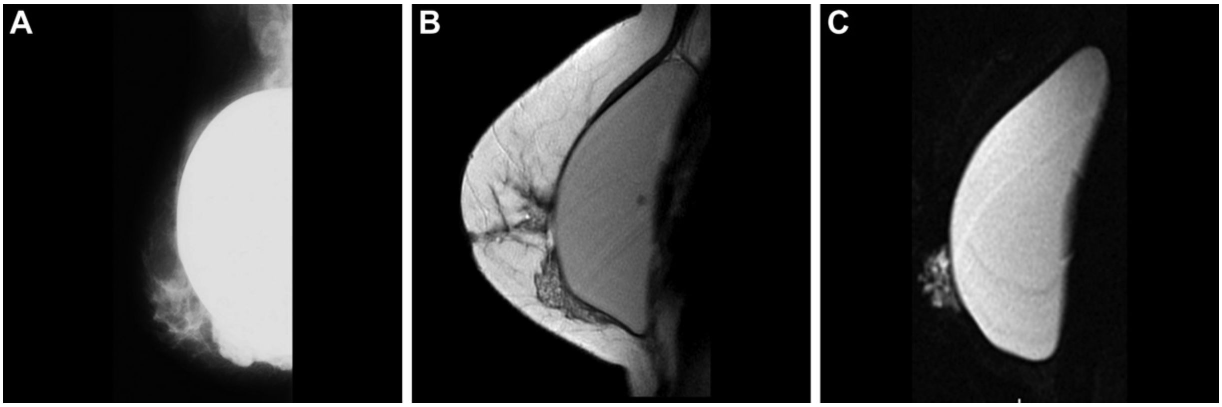


Figure 18. Residual silicone in the right breast tissue in a 40-year-old woman who underwent replacement of silicone implants in the past. **(A)** Right mediolateral-oblique projection MR image shows free silicone around the right breast implant and the adjacent breast parenchyma. **(B)** Sagittal T2-weighted MR image shows free silicone around the right retropectoral breast implant and the adjacent breast parenchyma. **(C)** Sagittal silicone-selective MR image confirms the presence of free silicone. There are no signs of intracapsular rupture.

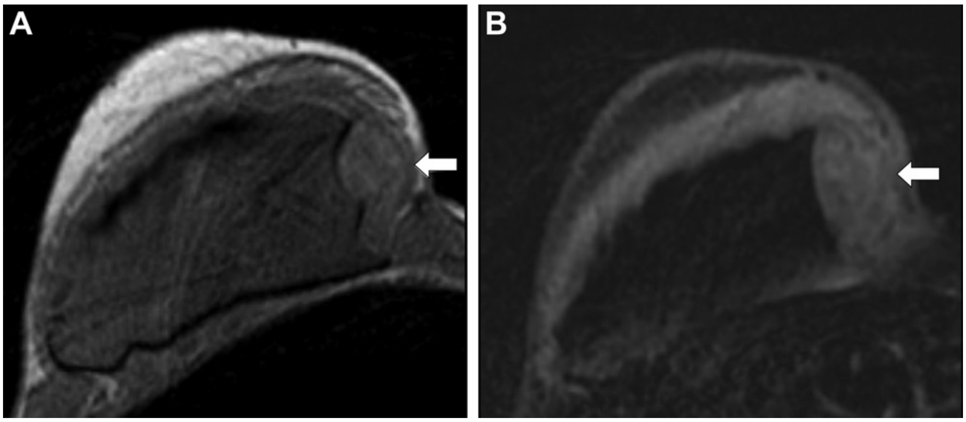


Figure 19. Palpable abnormality along the inner quadrant of the right breast in a 48-year-old woman with a history of bilateral mastectomy and reconstruction with implants. Doppler US image of the inner quadrant of the right breast showed a hypoechoic circumscribed mass with no vascularity and posterior acoustic shadowing (not shown). **(A)** Axial T1-weighted MR image shows an elongated mass (arrow) in the medial region of the right breast adjacent to the implant. **(B)** Axial contrast-enhanced T1-weighted MR image shows no enhancement of the mass (arrow). This mass corresponded to the location of a rolled acellular dermal matrix hammock or sling that was mentioned in the surgical description.

wall invasion. The appearance of these tumors is variable, from benign-appearing masses with circumscribed margins to irregular tumors with spiculated margins. They are usually isointense in comparison with muscle at T1-weighted MRI, with variable signal intensity at T2-weighted MRI, depending on the quantity of collagen and myxoid stroma. Bandlike hypointensities at T1- and T2-weighted MRI have been described (64).

Breast Implant–associated Lymphoma

BIA-ALCL is a rare T-cell lymphoma (ie, non-Hodgkin lymphoma) with a delayed manifestation. Patients usually present approximately 10 years after implantation; however, a new effusion that arises more than 1 year after implantation should raise suspicion for BIA-ALCL. It is a rare and relatively new entity, and the risk of BIA-ALCL is between one in 500 000 and one in 3 000 000 patients. The pathogenesis is yet unknown (22,65,66).

BIA-ALCL is seen more commonly with textured-surface implants than with smooth-surface implants. Patients present with breast swelling, pain, palpable masses, asymmetry, or lymphadenopathy (24,65). Systemic symptoms of BIA ALCL are rare. The initial workup for BIA-ALCL is US. Breast MRI is reserved for equivocal cases (Fig 23). When BIA-ALCL is suspected, breast MRI with intravenous contrast material is indicated to evaluate the peri-implant fluid, peri-implant masses, and capsular enhancement (7,22,23,58). The most common imaging finding is peri-implant fluid, followed by a soft-tissue mass. The current recommendation for a delayed peri-implant fluid collection

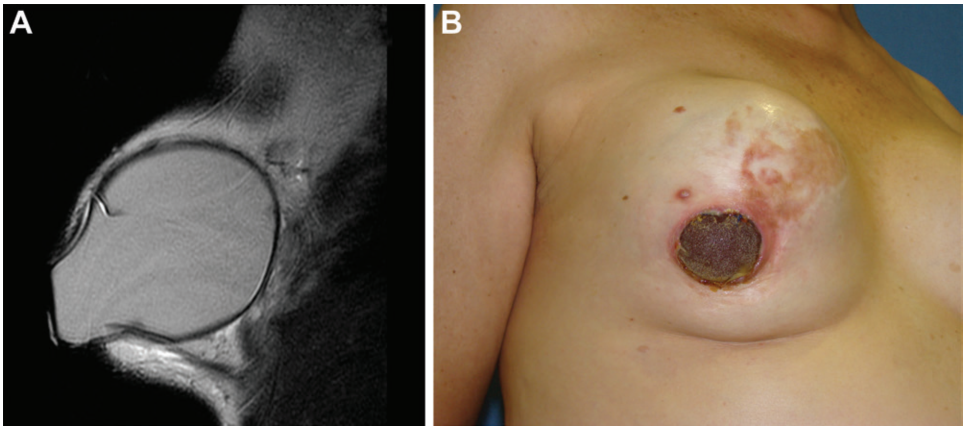


Figure 20. Capsular herniation with ulceration and infection of the skin in a 36-year-old woman with right breast pain. **(A)** Sagittal T2-weighted MR image of the right breast shows a silicone implant bulge in the inferolateral aspect of the implant contour. Herniation extends further outside the skin, causing ulceration and infection. **(B)** Photograph shows the ulceration and infection of the skin. The patient underwent removal of the right implant.

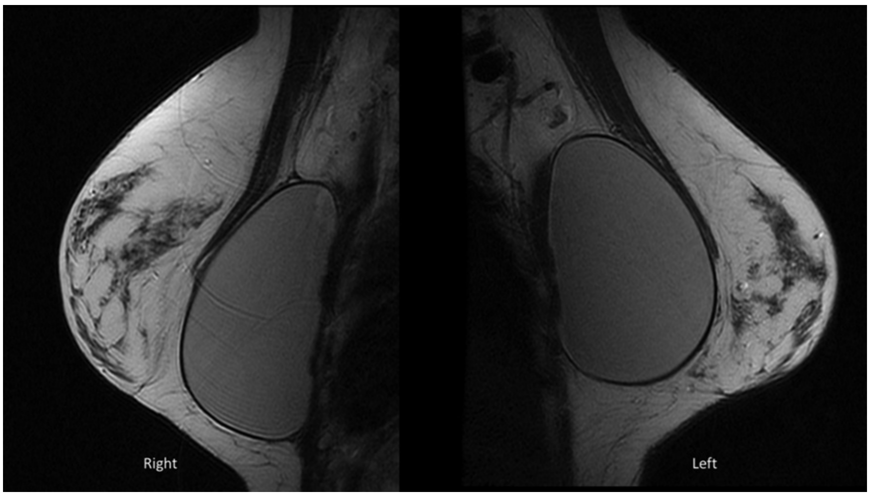
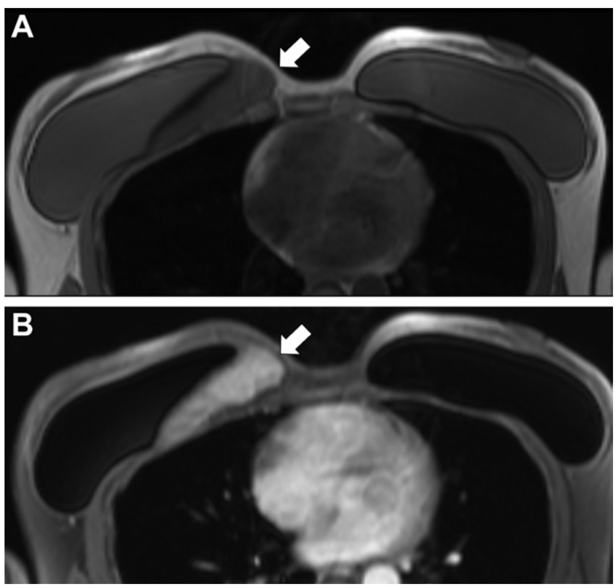


Figure 21. Implant displacement in a 37-year-old woman. Bilateral sagittal T2-weighted MR images show bilateral retropectoral implants, with inferior displacement of the right implant.

Figure 22. Desmoid tumor in a 36-year-old woman with right breast pain and a palpable abnormality in the right breast. US showed a circumscribed mass in the right breast, with internal vascularity on color Doppler US images (not shown). **(A)** Axial T1-weighted MR image shows an oval, circumscribed, isointense mass located in the submuscular area behind the right prosthesis (arrow). **(B)** Axial contrast-enhanced T1-weighted MR image shows the homogeneously enhancing mass (arrow).



is image-guided pathologic analysis with flow cytometry (24,65).

Fat Necrosis

Fat necrosis is a benign inflammatory process that occurs as a result of vascular compromise to adipose tissue, usually because of trauma or surgery (67). Fat necrosis is a frequent finding in patients who have undergone breast augmentation, reconstructive breast surgery, or autologous fat injections. Although it may manifest with

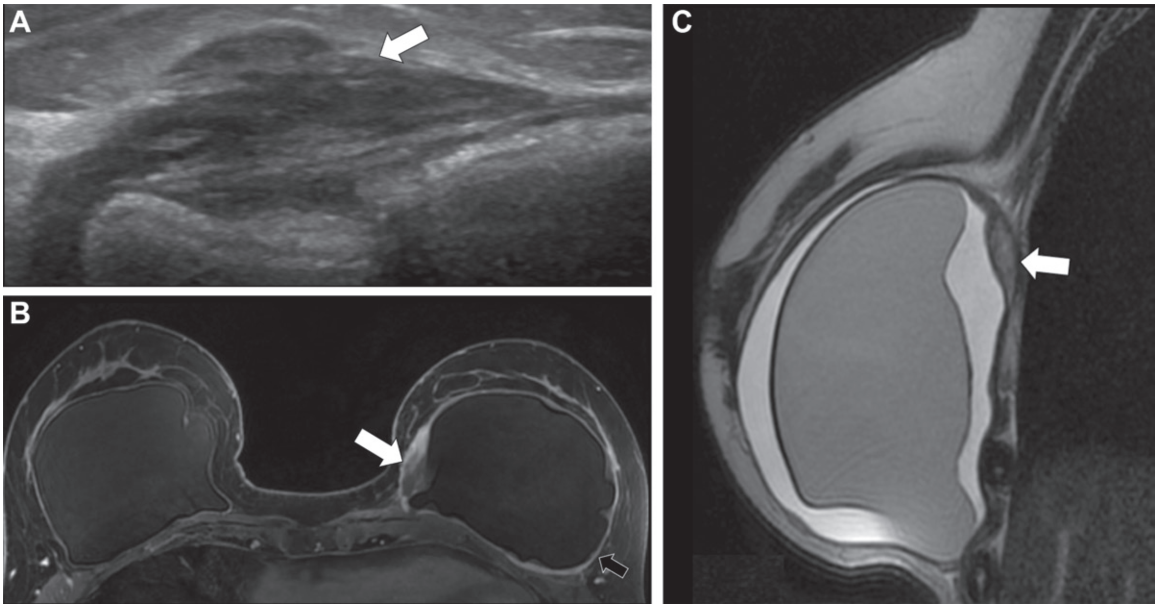


Figure 23. BIA-ALCL in a 62-year-old woman who presented with a palpable mass in the medial aspect of the left breast. **(A)** US image shows a solid mass (arrow) attached to the implant, correlating with the palpable mass. **(B)** Axial T1-weighted contrast-enhanced fat-saturation MR image shows the mass along the medial implant shell (white arrow), with rim enhancement. The implant capsule is thickened and enhances (black arrow). Biopsy results confirmed the diagnosis of BIA-ALCL. **(C)** Sagittal T2-weighted MR image of the right breast shows a peri-implant fluid collection and a hypointense mass posterior to the retropectoral implant (arrow). Immunohistochemical results (not shown) revealed CD30+ T-cell BIA-ALCL.

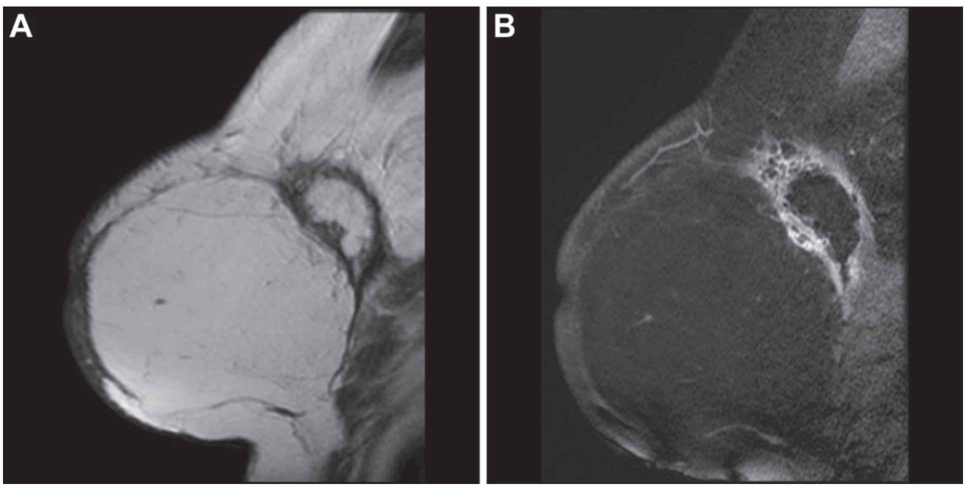


Figure 24. Fat necrosis in a 43-year-old woman with a history of cancer of the left breast who underwent mastectomy and reconstruction with a muscle-sparing free transverse rectus abdominus muscle flap. Sagittal T1-weighted **(A)** and sagittal contrast-enhanced T1-weighted fat-saturation **(B)** MR images of the left breast show a fat-containing mass in the upper quadrants with peripheral enhancement, which is a finding compatible with fat necrosis.

characteristically benign imaging features, fat necrosis may have imaging features that mimic breast cancer (15,24,43,68). The usual appearance is a round or oval lesion with high signal intensity at T1-weighted MRI and low signal intensity at fat-saturated MRI. Fat-fluid levels and rim enhancement have been described (Fig 24). These findings may persist for years and vary according to the stage of the lesion and the presence of inflammation and fibrosis. Alternatively, fat necrosis may manifest with features such as irregular masses or enhancing masses with spiculated margins at breast MRI that are indistinguishable from breast cancer. Biopsy is warranted to exclude malignancy. Mammography can be helpful to make a diagnosis when the characteristic mammographic appearance is present (67,69,70) (Fig 25).

Silicone Injection

Although it is performed less frequently, direct injection of silicone gel, liquid silicone, or biopolymers is used for breast augmentation. However,

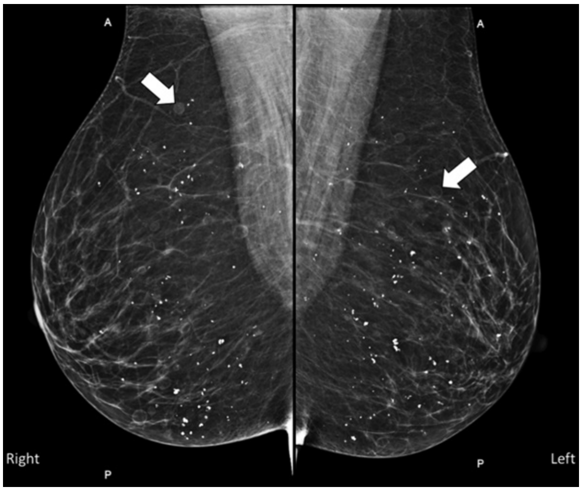


Figure 25. Oil cysts after breast augmentation with breast lipotropic injection in a 35-year-old woman. Bilateral mediolateral-oblique mammograms show multiple bilateral nodules, some of which developed mural calcifications.

the FDA has banned silicone injections. The silicone is either directly injected into the breast or the pectoralis major muscle (43) (Figs 26, 27). Because of the high density of this free injected material and the silicone granulomas with which it is associated, silicone injection markedly reduces the sensitivity of mammography. Therefore, breast MRI is a good option to detect breast cancer in this patient population. The imaging features are multiple T1-hypointense and T2-hyperintense masses at breast MRI. These masses also demonstrate high signal intensity at silicone-sensitive MRI. Complications of breast injection include granulomas, fibrosis, fat necrosis, lymphadenopathy, and occasionally, migration of the silicone to distant sites. Silicone granulomas may or may not exhibit enhancement (24,42,50,53,55,57).

Polyacrylamide gel injections are hypointense masses at T1-weighted and silicone-specific MRI and are hyperintense at T2-weighted MRI, with variable peripheral contrast enhancement (23,55).

Autologous fat injections manifest as masses with high signal intensity at non–fat-saturated T1-weighted MRI, with variable peripheral enhancement (55). Fat necrosis is more commonly seen when a higher amount of fat is injected or when injection occurs after radiation therapy (50,67).

Breast Cancer in Patients with Implants

The sensitivity of mammography is lower in patients with breast implants. This can be attributed to the difficulty in positioning patients, with the inevitable exclusion of some of the breast parenchyma during mammography. The presence of implants may delay detection of breast cancer (10,24,35,37). Handel et al (54) showed no significant differences in disease stage, tumor size, recurrence rates, or overall survival between patients with and those without breast implants. Moreover, small tumors may be easier to palpate in an augmented breast. A risk of prosthetic rupture during breast biopsy exists, and the patient's consent should always include statement of this risk (53).

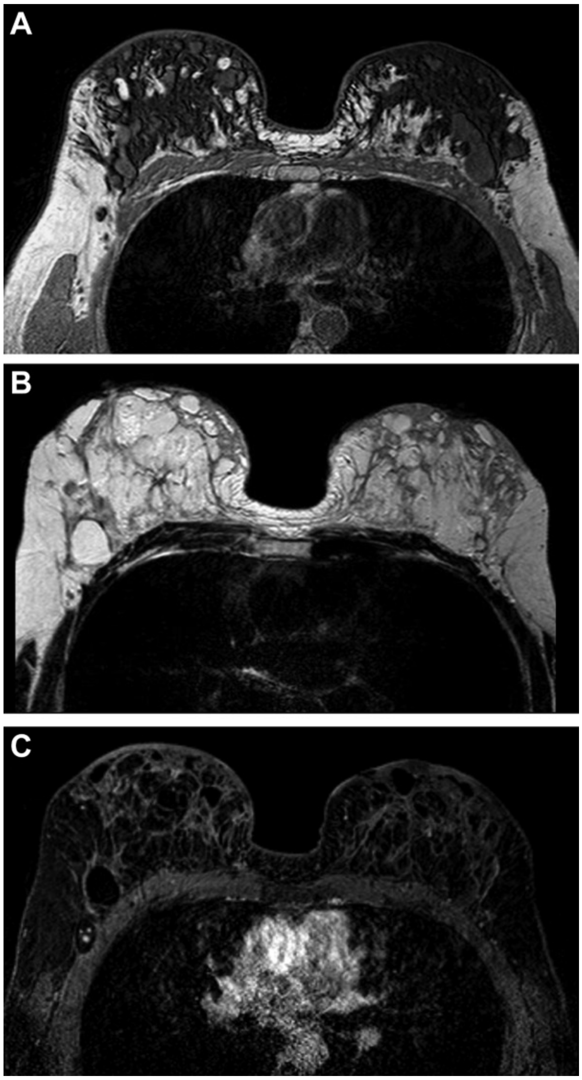


Figure 26. Biopolymer materials injected 3 years earlier to increase volume in the breast tissue in a 32-year-old woman. Axial T1-weighted **(A)**, T2-weighted **(B)**, and contrast-enhanced T1-weighted fat-saturation **(C)** MR images show multiple nodules in the breast tissue. The nodules are T1 hypointense **(A)** and T2 hyperintense **(B)** and show heterogeneous peripheral enhancement **(C)**.

Conclusion

Breast augmentation is one of most popular cosmetic surgical procedures worldwide. It is important for the radiologist to be familiar with the different methods of breast augmentation and the associated complications. Evaluating breast implants can be

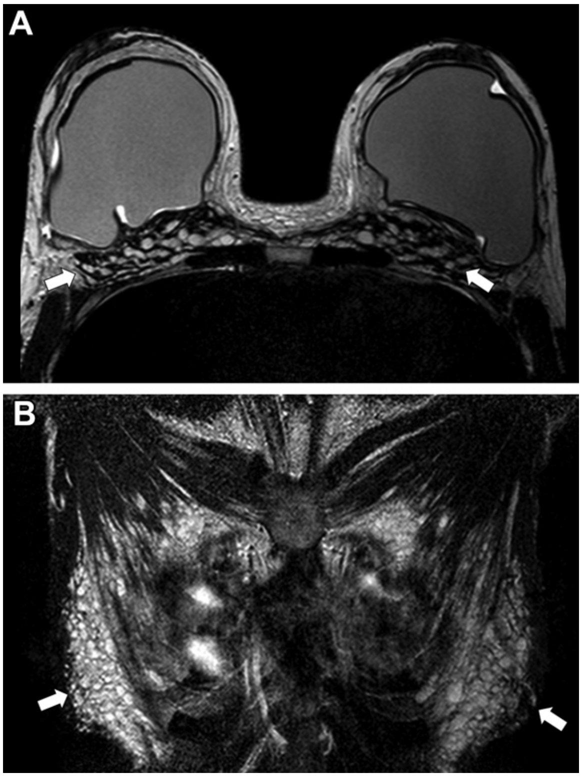


Figure 27. Bilateral breast implants in a 43-year-old woman with a history of injection of biopolymer material. T2-weighted **(A)** and fat-saturated T2-weighted **(B)** MR images show bilateral silicone single-lumen implants and multiple nodules in the chest wall infiltrating the breast tissue and the pectoral muscles (arrows).

challenging, but breast MRI is the imaging modality of choice to evaluate implant integrity and potential complications related to breast augmentation when the findings of conventional imaging are equivocal.

Acknowledgment.—The authors wish to acknowledge Mateo González González, MD, for creating the illustrations in Figures 3 and 12.

References

1. American Society of Plastic Surgeons. Plastic Surgery Statistics Report 2019. 2019 Plastic Surgery Statistics. 2019. p. 1–25. https://www.plasticsurgery.org/news/plastic-surgery-statistics?sub=2019+Plastic+Surgery+Statistics. Published 2019. Accessed March 1, 2022.
2. Kawahara S, Hyakusoku H, Ogawa R, Ohkubo S, Igarashi H, Hirakawa K. Clinical imaging diagnosis of implant materials for breast augmentation. Ann Plast Surg 2006;57(1):6–12.
3. Champaneria MC, Wong WW, Hill ME, Gupta SC. The evolution of breast reconstruction: a historical perspective. World J Surg 2012;36(4):730–742.
4. Bartlett W. An anatomic Substitute for the Female Breast. Ann Surg 1917;66(2):208–211.
5. Goldwyn RM. Vincenz Czerny and the beginnings of breast reconstruction. Plast Reconstr Surg 1978;61(5):673–681.
6. Shah AT, Jankharia BB. Imaging of common breast implants and implant-related complications: A pictorial essay. Indian J Radiol Imaging 2016;26(2):216–225.
7. Sardanelli F, Boetes C, Borisch B, et al. Magnetic resonance imaging of the breast: recommendations from the EUSOMA working group. Eur J Cancer 2010;46(8):1296–1316.
8. Middleton MS, McNamara MP Jr. Breast implant classification with MR imaging correlation. RadioGraphics 2000;20(3):E1. doi: 10.1148/radiographics.20.3.g00mae11.
9. Amano Y, Aoki R, Kumita S, Kumazaki T. Silicone-selective multishot echo-planar imaging for rapid MRI survey of breast implants. Eur Radiol 2007;17(7):1875–1878.
10. Herborn CU, Marincek B, Erfmann D, et al. Breast augmentation and reconstructive surgery: MR imaging of implant rupture and malignancy. Eur Radiol 2002;12(9):2198–2206.
11. Hölmich LR, Vejborg I, Conrad C, Sletting S, McLaughlin JK. The diagnosis of breast implant rupture: MRI findings compared with findings at explantation. Eur J Radiol 2005;53(2):213–225.
12. Tark KC, Jeong HS, Roh TS, Choi JW. Analysis of 30 breast implant rupture cases. Aesthetic Plast Surg 2005;29(6):460–469; discussion 470–471.
13. Seiler SJ, Sharma PB, Hayes JC, et al. Multimodality imaging-based evaluation of single-lumen silicone breast implants for rupture. RadioGraphics 2017;37(2):366–382.
14. Middleton MS. MR evaluation of breast implants. Radiol Clin North Am 2014;52(3):591–608.
15. Glynn C, Litherland J. Imaging breast augmentation and reconstruction. Br J Radiol 2008;81(967):587–595.
16. Juanpere S, Perez E, Huc O, Motos N, Pont J, Pedraza S. Imaging of breast implants-a pictorial review. Insights Imaging 2011;2(6):653–670.
17. Roller R, Chetlen A, Kasales C. Imaging of Breast Implants and Their Associated Complications. J Am Osteopath Coll Radiol 2014;3(1):2–9.
18. Brown SL, Middleton MS, Berg WA, Soo MS, Pennello G. Prevalence of rupture of silicone gel breast implants revealed on MR imaging in a population of women in Birmingham, Alabama. AJR Am J Roentgenol 2000;175(4):1057–1064.
19. Coroneos CJ, Selber JC, Offodile AC 2nd, Butler CE, Clemens MWUS. US FDA Breast Implant Postapproval Studies: Long-term Outcomes in 99,993 Patients. Ann Surg 2019;269(1):30–36.
20. Expert Panel on Breast Imaging; Lourenco AP, Moy L, et al. ACR Appropriateness Criteria® Breast Implant Evaluation. J Am Coll Radiol 2018;15(5S):S13–S25.
21. U.S. Food and Drug Administration. Saline, silicone gel, and alternative breast implants: guidance for industry and Food and Drug Administration Staff. https://www.fda.gov/regulatory-information/search-fda-guidance-documents/saline-silicone-gel-and-alternative-breast-implants. Published 2006. Updated 2020. Accessed March 1, 2022.
22. Wiedenhoefer JF, Shahid H, Dornbluth C, Otto P, Kist K. MR imaging of breast implants: Useful information for the interpreting radiologist. Appl Radiol 2015;(October):18–24. https://www.appliedradiology.com/articles/mr-imaging-of-breast-implants-useful-information-for-the-interpreting-radiologist. Accessed March 1, 2022.
23. Shah M, Tanna N, Margolies L. Magnetic resonance imaging of breast implants. Top Magn Reson Imaging 2014;23(6):345–353.
24. Green LA, Karow JA, Toman JE, Lostumbo A, Xie K. Review of breast augmentation and reconstruction for the radiologist with emphasis on MRI. Clin Imaging 2018;47:101–117.
25. American College of Radiology BI-RADS Committee. ACR BI-RADS Atlas. 5th ed. Reston, Va: American College of Radiology, 2013.
26. Schmitt W, Coelho JM, Lopes J, Marques JC. The Role of Radiology in Detecting Prosthetic Breast Implant-Related Complications. Acta Radiol Port 2018;30(1):23–34.
27. Barnsley GP, Sigurdson LJ, Barnsley SE. Textured surface breast implants in the prevention of capsular contracture among breast augmentation patients: a meta-analysis of randomized controlled trials. Plast Reconstr Surg 2006;117(7):2182–2190.
28. Morgan DE, Kenney PJ, Meeks MC, Pile NS. MR imaging of breast implants and their complications. AJR Am J Roentgenol 1996;167(5):1271–1275.
29. Lista F, Tutino R, Khan A, Ahmad J. Subglandular breast augmentation with textured, anatomic, cohesive silicone implants: a review of 440 consecutive patients. Plast Reconstr Surg 2013;132(2):295–303.

30. Biggs TM, Yarish RS. Augmentation mammaplasty: retropectoral versus retromammary implantation. Clin Plast Surg 1988;15(4):549–555.
31. Handel N, Silverstein MJ, Gamagami P, Jensen JA, Collins A. Factors affecting mammographic visualization of the breast after augmentation mammaplasty. JAMA 1992;268(14):1913–1917.
32. Ikeda DM, Borofsky HB, Herfkens RJ, Sawyer-Glover AM, Birdwell RL, Glover GH. Silicone breast implant rupture: pitfalls of magnetic resonance imaging and relative efficacies of magnetic resonance, mammography, and ultrasound. Plast Reconstr Surg 1999;104(7):2054–2062.
33. Hölmich LR, Vejborg IM, Conrad C, et al. Untreated silicone breast implant rupture. Plast Reconstr Surg 2004;114(1):204–214; discussion 215–216.
34. Gorczyca DP, Gorczyca SM, Gorczyca KL. The diagnosis of silicone breast implant rupture. Plast Reconstr Surg 2007;120(7 Suppl 1):49S–61S.
35. Topping A, George C, Wilson G. Appropriateness of MRI scanning in the detection of ruptured implants used for breast reconstruction. Br J Plast Surg 2003;56(2):186–189.
36. Brenner RJ. Evaluation of breast silicone implants. Magn Reson Imaging Clin N Am 2013;21(3):547–560.
37. Mann RM, Kuhl CK, Kinkel K, Boetes C. Breast MRI: guidelines from the European Society of Breast Imaging. Eur Radiol 2008;18(7):1307–1318.
38. Rudolph R, Forcier N. Calculation of silicone breast implant volumes using breast magnetic resonance imaging. Aesthet Surg J 2009;29(4):310–313.
39. Moschetta M, Telegrafo M, Capuano G, et al. Intraprosthetic breast MR virtual navigation: a preliminary study for a new evaluation of silicone breast implants. Magn Reson Imaging 2013;31(8):1292–1297.
40. Pfleiderer B, Heindel W. MRI and MR spectroscopy after silicone breast implants in the female breast [in German]. Rofo 2001;173(7):571–579.
41. Soo MS, Kornguth PJ, Walsh R, Elenberger CD, Georgiade GS. Complex radial folds versus subtle signs of intracapsular rupture of breast implants: MR findings with surgical correlation. AJR Am J Roentgenol 1996;166(6):1421–1427.
42. Ganott MA, Harris KM, Ilkhanipour ZS, Costa-Greco MA. Augmentation mammoplasty: normal and abnormal findings with mammography and US. RadioGraphics 1992;12(2):281–295.
43. Yang N, Muradali D. The augmented breast: a pictorial review of the abnormal and unusual. AJR Am J Roentgenol 2011;196(4):W451–W460.
44. Oliveira ACP, Maino M, Zanin EM, de Carli L, Duarte DW, Collares MVM. Breast Implants Follow-up: Results of a Cross-Sectional Study on Patients Submitted to MRI Breast Examinations. Aesthetic Plast Surg 2021;45(1):27–34.
45. Middleton MS. Magnetic resonance evaluation of breast implants and soft-tissue silicone. Top Magn Reson Imaging 1998;9(2):92–137.
46. O'Toole M, Caskey CI. Imaging spectrum of breast implant complications: mammography, ultrasound, and magnetic resonance imaging. Semin Ultrasound CT MR 2000;21(5):351–361.
47. Wong T, Lo LW, Fung PYE, et al. Magnetic resonance imaging of breast augmentation: a pictorial review. Insights Imaging 2016;7(3):399–410.
48. Mazzocchi M, Dessy LA, Carlesimo B, Marchetti F, Scuderi N. Late seroma formation after breast surgery with textured silicone implants: a problem worth bearing in mind. Plast Reconstr Surg 2010;125(4):176e–177e.
49. Oliveira VM, Roveda Junior D, Lucas FBP, et al. Late seroma after breast augmentation with silicone prostheses: a case report. Breast J 2007;13(4):421–423.
50. Yahyavi-Firouz-Abadi N, Menias CO, Bhalla S, Siegel C, Gayer G, Katz DS. Imaging of cosmetic plastic procedures and implants in the body and their potential complications. AJR Am J Roentgenol 2015;204(4):707–715.
51. Namnoum JD, Largent J, Kaplan HM, Oefelein MG, Brown MH. Primary breast augmentation clinical trial outcomes stratified by surgical incision, anatomical placement and implant device type. J Plast Reconstr Aesthet Surg 2013;66(9):1165–1172.
52. Behranwala KA, Dua RS, Ross GM, Ward A, A'hern R, Gui GPH. The influence of radiotherapy on capsule formation and aesthetic outcome after immediate breast reconstruction using biodimensional anatomical expander implants. J Plast Reconstr Aesthet Surg 2006;59(10):1043–1051.
53. Pelosi MA 3rd, Pelosi MA 2nd. Breast augmentation. Obstet Gynecol Clin North Am 2010;37(4):533–546, viii.
54. Handel N. The effect of silicone implants on the diagnosis, prognosis, and treatment of breast cancer. Plast Reconstr Surg 2007;120(7 Suppl 1):81S–93S.
55. Venkataraman S, Hines N, Slanetz PJ. Challenges in mammography: part 2, multimodality review of breast augmentation--imaging findings and complications. AJR Am J Roentgenol 2011;197(6):W1031–W1045.
56. Benito-Ruiz J, de Cabo F. Ultrasonography: a useful tool for plastic surgeons. Aesthetic Plast Surg 2014;38(3):561–571.
57. Scaranelo AM, Marques AF, Smialowski EB, Lederman HM. Evaluation of the rupture of silicone breast implants by mammography, ultrasonography and magnetic resonance imaging in asymptomatic patients: correlation with surgical findings. Sao Paulo Med J 2004;122(2):41–47.
58. Stöblen F, Rezai M, Kümmel S. Imaging in patients with breast implants—results of the First International Breast (Implant) Conference 2009. Insights Imaging 2010;1(2):93–97.
59. Cher DJ, Conwell JA, Mandel JS. MRI for detecting silicone breast implant rupture: meta-analysis and implications. Ann Plast Surg 2001;47(4):367–380.
60. Maijers MC, Niessen FB, Veldhuizen JFH, Ritt MJPF, Manoliu RA. MRI screening for silicone breast implant rupture: accuracy, inter- and intraobserver variability using explantation results as reference standard. Eur Radiol 2014;24(6):1167–1175.
61. Berg WA, Nguyen TK, Middleton MS, Soo MS, Pennello G, Brown SL. MR imaging of extracapsular silicone from breast implants: diagnostic pitfalls. AJR Am J Roentgenol 2002;178(2):465–472.
62. Vestito A, Mangieri FF, Ancona A, Minervini C, Perchinunno V, Rinaldi S. Study of breast implant rupture: MRI versus surgical findings. Radiol Med (Torino) 2012;117(6):1004–1018.
63. Dashevsky BZ, Hayward JH, Woodard GA, Joe BN, Lee AY. Utility and outcomes of imaging evaluation for palpable lumps in the postmastectomy patient. AJR Am J Roentgenol 2019;213(2):464–472.
64. Seo YN, Park YM, Yoon HK, Lee SJ, Choo HJ, Ryu JH. Breast fibromatosis associated with breast implants. Jpn J Radiol 2015;33(9):591–597.
65. Adrada BE, Miranda RN, Rauch GM, et al. Breast implant-associated anaplastic large cell lymphoma: sensitivity, specificity, and findings of imaging studies in 44 patients. Breast Cancer Res Treat 2014;147(1):1–14.
66. Ye X, Shokrollahi K, Rozen WM, et al. Anaplastic large cell lymphoma (ALCL) and breast implants: breaking down the evidence. Mutat Res Rev Mutat Res 2014;762:123–132.
67. Daly CP, Jaeger B, Sill DS. Variable appearances of fat necrosis on breast MRI. AJR Am J Roentgenol 2008;191(5):1374–1380.
68. Turner A, Abu-Ghname A, Davis MJ, Winocour SJ, Hanson SE, Chu CK. Fat grafting in breast reconstruction. Semin Plast Surg 2020;34(1):17–23.
69. Adrada BE, Whitman GJ, Crosby MA, Carkaci S, Dryden MJ, Dogan BE. Multimodality Imaging of the Reconstructed Breast. Curr Probl Diagn Radiol 2015;44(6):487–495.
70. Tayyab SJ, Adrada BE, Rauch GM, Yang WT. A pictorial review: multimodality imaging of benign and suspicious features of fat necrosis in the breast. Br J Radiol 2018;91(1092):20180213.